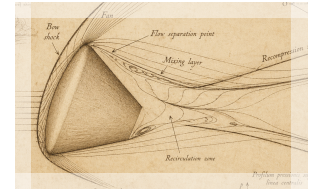


HyperStructures 2026

Continuum–Kinetic Models in Hypersonic Transitional Flows



📍 University of Nottingham, Jubilee Campus

📅 2–4 Sep 2026

👥 In-person

🔧 Workshop Overview

HyperStructures 2026 is a focused workshop examining modelling and validation challenges in transitional and rarefied hypersonic flow regimes. The meeting will bring together researchers from academia and industry to discuss the mathematical structure, regime validity, and numerical behaviour of models across the continuum–kinetic spectrum.

🔧 Modelling Spectrum

- Navier–Stokes and breakdown regimes
- Burnett and higher-order continuum models
- Thermochemistry
- DSMC and kinetic–continuum interfaces
- Hybrid and emerging formulations

⚙️ Core Themes

- Closure and stability
- Hyperbolicity and entropy structure
- Continuum-limit consistency
- Thermochemistry
- Numerical resolution and robustness
- Validation and benchmarking across regimes

💬 Format

The workshop will be discussion-oriented rather than lecture-based, with short framing talks followed by structured sessions aimed at identifying open modelling and validation challenges in hypersonic transitional regimes.

💡 Motivation

Transitional hypersonic regimes expose fundamental limitations of classical continuum models while remaining computationally demanding for fully kinetic approaches. Clarifying structural assumptions, regime validity, and continuum–kinetic consistency is essential for reliable modelling and validation.

✉️ Contact

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Registration / travel
form